INE4307 — Introduction to Decision Analysis

Midterm Study Pack · 60 True/False · 40 Multiple Choice · 10 Calculator Problems · 2 Worked Comprehensive Cases

Exam structure (100 points): Q1 True/False, 10 statements — 20 pts | Q2 Multiple choice, 10 questions — 30 pts | Q3 Comprehensive problem, parts (a)–(e) — 50 pts
Q3 chain: payoff matrix -> influence diagram -> decision tree with backward induction -> cumulative risk profiles + dominance test -> break-even sensitivity analysis.
Scope: W1–W3. Bayes' theorem, value of information, cognitive biases and utility theory are W4–W6 (EVPI appears only as an optional extension).
Three marks-losing details: mark pruned branches with **//** · the cumulative profile is **F(x) = P(Payoff <= x)**, not P(>=) · "influential on the value" is not the same as "influential on the decision" — only the second belongs in a requisite model.

Part 1 — 60 True / False Statements

#	Statement	Ans	Why
Structuring and Elements			
1	Decision analysis is a **prescriptive** approach.	T	Keeney & Raiffa definition.
2	The main output of decision analysis is a guaranteed correct solution.	F	It provides **insight**, not necessarily a solution.
3	A requisite decision model is one whose form and content are **just sufficient** to solve the problem.	T	Sensitivity analysis is how you find out what is sufficient.
4	To move from a means objective towards a fundamental objective you ask "Why is that important?"	T	WITI.
5	To move **down** a fundamental objectives hierarchy you ask "What do you mean by that?"	T	Upward: "Of what more general objective is this an aspect?"
6	Fundamental objectives are organised into hierarchies; means objectives into networks.	T	Standard distinction.
7	A consequence is a function of the decision only.	F	It is a function of the decision **and** the state of nature.
8	"Doing nothing" should never be treated as an alternative.	F	Always consider doing nothing, waiting, or hedging.
9	Only uncertain events that affect at least one objective are relevant to the model.	T	Relevance criterion.
10	The planning horizon is the end of the time line looking forward from the current decision.	T	Definition.
11	Decision analysis can only be applied when consequences are monetary.	F	Multiple objectives, medicine, public policy, law.
12	A payoff table is one way of representing a decision problem.	T	Alternatives by states of nature.
Influence Diagrams			
13	Rectangle = decision node, oval = chance node, diamond = payoff node.	T	Rounded rectangle = intermediate calculation.
14	A rounded rectangle represents an intermediate calculation node.	T	Aggregates predecessors for clarity.
15	An arc pointing **into a decision node** represents **sequence**.	T	The predecessor is known before the decision.
16	An arc pointing **into a chance node** represents **relevance**.	T	It affects the probability assessment.
17	An arc pointing into a value or calculation node is **functional**.	T	It enters the value computation.
18	An influence diagram may contain a cycle.	F	It must **never** contain cycles.
19	An arc may leave the payoff (value) node.	F	The value node is terminal; it has no successors.
20	Clemen & Reilly use dashed arcs for sequence.	F	All arcs are solid; meaning follows from where they point.
21	If there is no arc from a chance node to a decision node, the outcome is unknown when the decision is made.	T	Decision under uncertainty, not under information.
22	An arc from a decision node to a chance node means the decision affects the probabilities.	T	Act-dependent probabilities.
23	An influence diagram shows more detail than the corresponding decision tree.	F	The **tree** shows more detail; the ID is compact.
24	An influence diagram and its decision tree are isomorphic.	T	Same problem, two representations.

#	Statement	Ans	Why
25	Influence diagrams are easier to explain to upper-level managers.	T	Compactness is the advantage.
26	Interpreting an influence diagram is harder than creating one.	F	The reverse: interpreting is easy, creating is hard.
27	An influence diagram is a snapshot of the current state of knowledge.	T	Definition.
28	Calculation nodes are useful when a node has many predecessors.	T	Defines relationships more precisely.
29	Every influence diagram must contain at least one calculation node.	F	Entirely optional.
30	Nodes in an influence diagram are defined by means of tables.	T	Conditional probability / value tables.
Decision Trees, EMV and Backward Induction			
31	A decision tree flows from left to right.	T	Time sequence.
32	In a decision tree a square is a decision node and a circle is a chance node.	T	Standard symbols.
33	Branches out of a chance node must be mutually exclusive and collectively exhaustive.	T	MECE.
34	Probabilities on branches out of a chance node must sum to 1.	T	Basic requirement.
35	Probabilities shown in a decision tree are unconditional.	F	They are conditional on events already observed to the left.
36	If a chance node lies to the right of a decision node, the decision is made in anticipation of the event.	T	The outcome is not yet known.
37	If a chance node lies to the left of a decision node, the decision is conditional on that outcome.	T	Information before the decision.
38	In folding back you compute the EMV at a decision node.	F	You take the **maximum** (the minimum when minimising cost).
39	When minimising cost you select the lowest EMV at a decision node.	T	Direction flips, method does not.
40	Pruned, non-optimal branches are marked with the symbol //.	T	Required notation in this course.
41	A decision path selects one alternative at each decision node and one outcome at each chance node.	T	A possible future scenario.
42	A decision strategy specifies what to do at every decision node that might be reached.	T	A collection of decision paths.
43	The optimal decision strategy gives the highest EMV when maximising profit.	T	And the lowest EMV when minimising cost.
44	More than one alternative may be selected at a single decision node.	F	Exactly one.
45	Cumulative monetary values are shown to the right of the end nodes.	T	Individual values sit on the branches.
Risk Profiles and Dominance			
46	A risk profile is a bar chart of the probabilities of the possible outcomes.	T	Continuous distributions can also be plotted.
47	Each risk profile is associated with one decision strategy.	T	Not with a single branch.
48	The cumulative risk profile is F(x) = P(Payoff <= x).	T	Not P(Payoff >= x).
49	Deterministic dominance means the worst payoff of A is at least the best payoff of B.	T	Ranges do not overlap.
50	If two cumulative risk profiles cross, stochastic dominance exists.	F	It does **not**; dominance cannot resolve the comparison.
51	First-degree stochastic dominance requires F_A(x) <= F_B(x) for every x.	T	And the curves must never cross.
52	A dominated alternative may be eliminated from further analysis.	T	That is the point of the test.
53	If dominance cannot resolve a comparison, the choice depends on risk attitude.	T	Maximin, EMV, or a utility function.
54	The EMV criterion accounts for the decision maker's risk attitude.	F	EMV is risk-neutral by construction.
55	A risk-averse decision maker may prefer an alternative with a lower EMV but no chance of loss.	T	Explicit risk modelling.
56	When you buy insurance the expected value always favours the insurance company.	T	Stated in the lecture.

#	Statement	Ans	Why
57	If there is no deterministic dominance there cannot be stochastic dominance either.	F	They are separate, successively weaker tests.
58	Second-degree stochastic dominance of A over B requires the cumulated area difference to stay non-negative.	T	G(t) = integral of (F_B - F_A) >= 0 for all t.
59	In a second-degree dominance test, G at plus infinity equals EMV(A) - EMV(B).	T	A free arithmetic check.
Sensitivity Analysis			
60	A tornado diagram varies every input, one at a time, for a single alternative.	T	Relative effect.
61	A sensitivity graph varies one input across its range and shows all alternatives.	T	Individual, exact effect.
62	Tornado bars are sorted by swing, widest at the top.	T	Hence the tornado shape.
63	You use the tornado diagram first and the sensitivity graph second.	T	They are complementary, not competing.
64	A tornado bar that runs backwards indicates an error.	F	It is information: a negative relationship.
65	One-way sensitivity analysis overestimates sensitivity.	F	It **underestimates** it; interactions are ignored.
66	Two-way sensitivity analysis partitions the plane into strategy regions.	T	Connected areas where one alternative wins.
67	Every variable with a large swing belongs in the requisite model.	F	Only if it changes the ranking of the alternatives.
68	With three mutually exclusive and exhaustive states the feasible probability region is a triangle.	T	t >= 0, v >= 0, t + v <= 1.
69	With three alternatives the three pairwise indifference lines meet at a single point.	T	Use it to check your algebra.
70	EMV is linear in the probabilities.	T	So indifference boundaries are straight lines.
71	A tornado diagram tells you which alternative wins.	F	The sensitivity graph does that.
72	There is a single optimal procedure for sensitivity analysis.	F	Model building, and hence sensitivity analysis, is an art.
73	An alternative whose payoff is identical in two states gives a flat line when probability shifts between them.	T	Shifting probability between equal payoffs cannot change the mean.

Part 2 — 40 Multiple Choice Questions

#	Question and options	Ans	Why
1	**Which shape represents a decision node in an influence diagram?** A) Oval B) **Rectangle** C) Diamond D) Circle	B	Oval = chance, diamond = payoff, rounded rectangle = calculation.
2	**An arc from a chance node to a decision node means:** A) The decision changes the probabilities B) **The outcome is known before the decision is made** C) The two nodes are independent D) A cycle has been created	B	Arcs into a decision node represent sequence.
3	**Two cumulative risk profiles cross. It follows that:** A) The one on the right dominates B) **There is no first-degree stochastic dominance** C) Both alternatives have equal EMV D) One alternative is deterministically dominated	B	Dominance cannot resolve the comparison; risk attitude decides.
4	**At a decision node during backward induction you:** A) Average the branch values B) **Select the best EMV and prune the rest** C) Multiply the branch probabilities D) Add the branch payoffs	B	Averaging out happens at chance nodes only.
5	**The top bar of a tornado diagram belongs to:** A) The most likely variable B) The variable with the smallest range C) **The variable with the largest swing** D) The decision variable	C	Bars are sorted by swing, widest at the top.
6	**Deterministic dominance of A over B means:** A) EMV(A) > EMV(B) B) **A's worst payoff is at least B's best payoff** C) F_A(x) <= F_B(x) for all x D) A has a smaller variance	B	The payoff ranges must not overlap.
7	**Which can never appear in an influence diagram?** A) A calculation node B) **A cycle** C) Two chance nodes in series D) An arc into the value node	B	Influence diagrams are acyclic by definition.
8	**If the probabilities are act-dependent, the influence diagram must contain:** A) **An arc from the decision node to the chance node** B) An arc from the chance node to the decision node C) A dashed arc D) Two value nodes	A	The act changes the probability distribution.
9	**A risk profile corresponds to:** A) A single branch B) A single state of nature C) **A decision strategy** D) The whole decision tree	C	One profile per strategy.
10	**Two-way sensitivity analysis is needed because one-way analysis:** A) Is too slow B) **Ignores additive and multiplicative interaction between two variables** C) Cannot handle probabilities D) Requires a spreadsheet	B	One-way therefore underestimates sensitivity.
11	**A requisite decision model is one that:** A) Includes every conceivable variable B) **Has just sufficient form and content to solve the problem** C) Uses only objective probabilities D) Always yields a unique answer	B	Sensitivity analysis identifies what is sufficient.
12	**The break-even probability between two alternatives is found by:** A) Averaging their payoffs B) **Setting their EMVs equal and solving for p** C) Taking the ratio of their variances D) Reading it off the tornado diagram	B	EMV is linear in p, so the solution is unique.

#	Question and options	Ans	Why
13	**The EMV criterion assumes the decision maker is:** A) Risk averse B) Risk seeking C) **Risk neutral** D) Ambiguity averse	C	Risk attitude requires utility theory.
14	**An arc pointing into a value node is:** A) A sequence arc B) A relevance arc C) **A functional arc** D) Not permitted	C	It enters the value computation.
15	**Which statement about influence diagrams and decision trees is correct?** A) The ID shows more detail B) They model different problems C) **They are isomorphic representations of the same problem** D) The tree cannot show sequence	C	Two views of one model.
16	**In the festival venue problem, the single most common error is drawing an arc:** A) From Weather to Profit B) **From Venue Decision to Weather** C) From Venue Decision to Profit D) From Weather to Venue Decision	B	Booking an indoor arena does not change the chance of rain.
17	**Probabilities are act-independent when:** A) They sum to one B) **$P(S_j \mid d_i) = P(S_j)$ for every i** C) Every alternative has the same payoff D) The states are equally likely	B	Then one row of column probabilities describes the whole table.
18	**State-by-state (cell-by-cell) dominance testing is valid only when:** A) Payoffs are positive B) **All alternatives face the same underlying uncertainty** C) There are exactly two states D) EMVs are equal	B	Otherwise the columns are not comparable.
19	**An alternative that pays the same amount in two states will appear on a sensitivity graph over those probabilities as:** A) A steep line B) **A horizontal line** C) A parabola D) Two separate segments	B	Moving probability between identical payoffs cannot change the mean.
20	**With $P(S_3)$ fixed and $p = P(S_1)$, the slope of EMV for an alternative equals:** A) Its payoff in S_1 B) **Its payoff in S_1 minus its payoff in S_2** C) Its EMV at p = 0 D) The sum of all its payoffs	B	Which is why a flat alternative has equal payoffs in S1 and S2.
21	**A break-even value that falls inside the region where a third alternative wins:** A) Invalidates the analysis B) **Never appears as a boundary of the strategy regions** C) Must be recomputed D) Indicates an arithmetic error	B	Same behaviour as the third indifference line in the stock market example.
22	**If every pair of cumulative curves crosses, the next step is to:** A) Declare the problem unsolvable B) Recompute the EMVs C) **Test second-degree stochastic dominance** D) Apply maximax	C	A weaker but often decisive test.
23	**Second-degree stochastic dominance of A over B requires:** A) F_A(x) <= F_B(x) everywhere B) **The cumulated area of (F_B - F_A) to stay non-negative** C) Equal means D) Non-overlapping ranges	B	G(t) >= 0 for every t.
24	**In a second-degree dominance test, G evaluated at the top of the range equals:** A) 1 B) 0 C) **EMV(A) - EMV(B)** D) The variance difference	C	Use it to check your integration.

#	Question and options	Ans	Why
25	**An alternative eliminated by second-degree dominance against every rival:** A) Should still be considered by a risk seeker B) **Would be chosen by no risk-neutral or risk-averse decision maker** C) Has the highest EMV D) Must have the widest range	B	It is discarded regardless of the degree of risk aversion.
26	**EVPI is defined as:** A) EVwPI + EMV* B) **EVwPI - EMV*** C) EMV* / EVwPI D) The largest payoff minus the smallest	B	The most one should pay for a perfect forecast.
27	**EVPI is large relative to the gap between the best two alternatives when:** A) The payoffs are large B) **Perfect information changes the chosen action in many states** C) The probabilities are equal D) The EMVs are close	B	Information is only worth having where it changes the action.
28	**An imperfect forecast is worth:** A) More than EVPI B) Exactly EVPI C) **Strictly less than EVPI** D) Nothing	C	EVPI is an upper bound.
29	**A decision under uncertainty rather than under information is shown by:** A) A dashed arc B) **No arc entering the decision node** C) A calculation node D) Two chance nodes	B	Nothing is observed before the decision.
30	**If two states of an alternative give the same payoff, its risk profile has:** A) Two identical bars B) **One bar whose probability is the sum of the two** C) No bars D) A negative bar	B	Merge the outcomes and add the probabilities.
31	**Maximin selects:** A) The largest EMV B) **The alternative with the best worst case** C) The alternative with the best best case D) The middle alternative	B	A pessimistic, non-probabilistic criterion.
32	**Maximax selects:** A) The best worst case B) **The alternative with the best best case** C) The largest EMV D) The smallest variance	B	It can tie and then fails to discriminate.
33	**A compromise (middle) alternative often wins on EMV because:** A) It has the highest ceiling B) It has the highest floor C) **It is strong in the middle state, which carries substantial probability** D) It has the smallest range	C	As in the festival venue problem.
34	**A tornado bar of zero width means:** A) The variable is certain B) **That variable has no effect on that particular alternative** C) The model is wrong D) The variable should be removed from the model	B	A property of the alternative, not of the model.
35	**Two tornado diagrams for two alternatives may rank the variables differently because:** A) One of them is wrong B) **A tornado describes a single alternative only** C) The bounds differ D) The base case differs	B	Astoria and Barnard give different orderings.
36	**A variable with a large swing that shifts all alternatives equally is:** A) Kept and estimated carefully B) **Influential on the value but not on the decision** C) Deterministically dominant D) The tornado's shortest bar	B	It does not belong in the requisite model.

#	Question and options	Ans	Why
37	**In sensitivity to probabilities with three states, the feasible region is:** A) The unit square B) **The triangle t >= 0, v >= 0, t + v <= 1** C) A circle D) The whole plane	B	The probabilities must be non-negative and sum to at most one.
38	**Three pairwise indifference lines for three alternatives:** A) Are always parallel B) **Meet at a single point** C) Never intersect D) Form a triangle of positive area	B	If yours do not concur, you have made an arithmetic error.
39	**An alternative optimal only in a very small strategy region tells the decision maker that:** A) The model is wrong B) **Refining the probability assessment is worth little** C) That alternative should be removed D) The EMVs are equal	B	The low-risk stock behaves this way in the stock market example.
40	**Which sequence matches the required structure of the comprehensive question?** A) Tree, matrix, diagram, dominance, sensitivity B) **Matrix, influence diagram, tree, cumulative profiles with dominance, break-even** C) Diagram, sensitivity, tree, matrix, dominance D) Matrix, tree, sensitivity, diagram, dominance	B	Follow this order and label each part.

Part 3 — 10 Calculator Problems with Solutions

Where the calculator actually earns its keep: (a) net payoffs after an upfront cost; (c) the sum of probability times value at every chance node, and products of branch probabilities in a sequential tree; (d) cumulative sums and tail probabilities; (e) break-even fractions converted to decimals; and swings if a tornado is asked for. Part (b), the influence diagram, needs no arithmetic at all — move through it quickly.
Error traps: the sign on negative payoffs · subtracting the upfront cost twice · writing P(>= x) when the cumulative profile is P(<= x).

Problem 1 — EMV of a three-state alternative
Payoffs +900 / +400 / -300 with probabilities 0.40 / 0.35 / 0.25.

Solution. 0.40(900) = 360
0.35(400) = 140
0.25(-300) = -75
EMV = 360 + 140 - 75 = 425.0
A negative payoff subtracts. This is the single most common arithmetic slip.

Problem 2 — Net payoff after an upfront cost
Revenue 500K if the machines perform well, 100K if they do not; upfront investment 200K.

Solution. Good: 500 - 200 = **+300K**
Poor: 100 - 200 = **-100K**
Subtract the cost once, at the endpoint. Do not subtract it again inside the EMV.

Problem 3 — Choosing among three alternatives
d1: 0.70 of 300 and 0.30 of -100. d2: 0.60 of 300 and 0.40 of 50. d3 (lease, cost 100K): 0.55 of 240 and 0.45 of 40.

Solution. EMV(d1) = 210 - 30 = **180**
EMV(d2) = 180 + 20 = **200**
EMV(d3) = 132 + 18 = **150**
Choose d2; prune d1 and d3 with //.

Problem 4 — Sequential tree and path probabilities
Run a market test costing 20 first. P(Favourable) = P(Unfavourable) = 0.5; P(Success | F) = 0.8, P(Success | U) = 0.2. Launching pays 400 on success, -200 on failure; abandoning pays 0.

Solution. If F: 0.8(400) + 0.2(-200) = 320 - 40 = **280** -> launch
If U: 0.2(400) + 0.8(-200) = 80 - 160 = **-80** -> abandon, take 0
Test node: 0.5(280) + 0.5(0) = 140; less the cost 20 -> **120**
Path probability of "favourable and success" = 0.5 x 0.8 = **0.40**
Optimal strategy: test; launch if favourable, abandon if unfavourable.

Problem 5 — Building a cumulative risk profile
A strategy yields +380 with probability 0.40, -220 with probability 0.10 and -20 with probability 0.50.

Solution. Sort ascending and cumulate:
F(-220) = 0.10
F(-20) = 0.10 + 0.50 = **0.60**
F(+380) = **1.00**
P(Payoff <= 0) = **0.60** P(Payoff >= 0) = 1 - 0.60 = **0.40**

Problem 6 — Break-even between two alternatives
EMV(A) = 500p + 225 and EMV(B) = 200p + 400.

Solution. 500p + 225 = 200p + 400
300p = 175
p = 0.5833
Choose A when p > 0.5833, otherwise B.

Problem 7 — Strategy regions with three alternatives
Add a third alternative with a constant EMV(C) = 437.5.

Solution. A = C: 500p + 225 = 437.5 -> 500p = 212.5 -> p = **0.4250**
B = C: 200p + 400 = 437.5 -> 200p = 37.5 -> p = **0.1875**
Regions: p < 0.1875 -> C | 0.1875 < p < 0.5833 -> B | p > 0.5833 -> A
0.4250 lies inside B's region, so it is never a boundary.

Problem 8 — Expected value of perfect information
Best payoff by state: 900 (p = 0.40), 500 (p = 0.35), 400 (p = 0.25). The best alternative without information has EMV* = 480.

Solution. EVwPI = 360 + 175 + 100 = **635**
EVPI = 635 - 480 = 155
Pay at most 155 for a perfectly reliable forecast; any real forecast is worth strictly less.

Problem 9 — Tornado swings and ranking
Base net annual cost 28,673. Interest rate moves it between 25,381 and 34,724; repair and upkeep between 27,273 and 36,273; tax rate between 30,412 (low) and 26,065 (high).

Solution. Interest rate: 34,724 - 25,381 = **9,343**
Repair and upkeep: 36,273 - 27,273 = **9,000**
Tax rate: 30,412 - 26,065 = **4,347**
Rank: interest rate, then repair, then tax rate. The tax rate bar runs **backwards** (the upper bound gives the lower cost, because a higher rate means a larger deduction).

Problem 10 — Sensitivity to probabilities, two variables
Net payoffs after the brokerage fee: high-risk stock 1500 / 100 / -1000, low-risk stock 1000 / 200 / -100, savings account 500 for certain. Let t = P(Up) and v = P(Same).

Solution. EMV(HRS) = 1500t + 100v - 1000(1 - t - v) = **2500t + 1100v - 1000**
EMV(LRS) = 1000t + 200v - 100(1 - t - v) = **1100t + 300v - 100**
At t = 0.50, v = 0.30: HRS = 1250 + 330 - 1000 = **580**; LRS = 550 + 90 - 100 = **540**; SA = **500** -> choose **HRS**
Boundary SA = LRS: 1100t + 300v = 600, so at v = 0, t = 600/1100 = **0.545**

Part 4 — Worked Comprehensive Case 1: The Festival Venue Decision

All payoffs in thousand TL. This is the in-class exercise the instructor told you to work through again.

Alternative	S1 Dry (0.40)	S2 Mixed (0.35)	S3 Heavy rain (0.25)	EMV
d1 Open-air waterfront	+900	+400	-300	425.0
d2 Waterfront + marquee	+700	+500	+100	480.0 (best)
d3 Indoor arena	+450	+450	+400	437.5

(a) Payoff matrix

No alternative is dominated state by state: each is best in exactly one column — d1 when dry, d2 when mixed, d3 in heavy rain. The probabilities must therefore be used.
The probabilities here are **act-independent**: $P(S_j \mid d_i) = P(S_j)$ for every i, so one row of column probabilities describes the whole table and cell-by-cell comparison is meaningful. *Contrast the factory machine problem, where each alternative carries its own private uncertainty and the probabilities are act-dependent.*

(b) Influence diagram — and the two arcs that must NOT be drawn

Nodes: **Venue Decision** (rectangle), **Weather at the Festival** (oval, outcomes S1/S2/S3 with 0.40/0.35/0.25), **Profit** (diamond, value).
Arcs present: Weather -> Profit (**functional**) and Venue Decision -> Profit (**functional**, through rental cost and capacity).
No arc from Venue Decision to Weather. That would be a relevance arc asserting that the choice of venue changes the weather distribution. Booking an indoor arena does not make rain more or less likely. This is the single most common error in the question.
No arc into Venue Decision. Nothing about the weather is observed before booking, so this is a decision under uncertainty, not under information.

(c) Decision tree and backward induction

EMV(d1) = 0.40(900) + 0.35(400) + 0.25(-300) = 360 + 140 - 75 = **425.0**
EMV(d2) = 0.40(700) + 0.35(500) + 0.25(100) = 280 + 175 + 25 = **480.0**
EMV(d3) = 0.40(450) + 0.35(450) + 0.25(400) = 180 + 157.5 + 100 = **437.5**
Fold back: EMV* = max{425.0, 480.0, 437.5} = **480.0** -> choose **d2, the waterfront site with a marquee**. Prune d1 and d3 with //.
The winner is the middle option: the marquee buys most of the weather protection of the arena while keeping most of the upside of the open site. Compromise alternatives are often optimal under EMV because they are strong in the middle state, which here carries 35 per cent of the probability.

(d) Cumulative risk profiles and dominance

Interval of x	F for d1	F for d2	F for d3
x < -300	0.00	0.00	0.00
-300 <= x < 100	0.25	0.00	0.00
100 <= x < 400	0.25	0.25	0.00
400 <= x < 450	0.60	0.25	0.25
450 <= x < 500	0.60	0.25	1.00
500 <= x < 700	0.60	0.60	1.00
700 <= x < 900	0.60	1.00	1.00
x >= 900	1.00	1.00	1.00

P(loss): d1 = **0.25**, d2 = 0, d3 = 0. Worst case: -300 / +100 / **+400**. Best case: **+900** / +700 / +450.
Deterministic dominance: none. The ranges [-300, 900], [100, 700] and [400, 450] all overlap; for instance min(d3) = 400 < max(d2) = 700.
First-degree stochastic dominance: none. Every pair of curves crosses — d2 against d1 at x = 700, d3 against d2 at x = 450, d3 against d1 at x = 450.
Second-degree stochastic dominance: d2 dominates d1 and d3 dominates d1, since the cumulated area difference G(t) stays non-negative throughout (and G at the top equals the EMV difference, 55 and 12.5 respectively — a free check). d2 and d3 do not dominate each other.
Recommendation. Risk neutral: d2, the highest EMV at 480.0 with no probability of loss. Risk averse: d2 or d3 — both remove

the loss entirely; d2 has the higher mean and a higher chance of clearing 450, while d3 has the far better floor (+400 against +100). Under any attitude, **never d1**: it is second-degree dominated by both rivals.

(e) Break-even sensitivity analysis

Hold P(S3) = 0.25 and let p = P(S1), so P(S2) = 0.75 - p:

EMV(d1) = 900p + 400(0.75 - p) - 300(0.25) = **500p + 225**

EMV(d2) = 700p + 500(0.75 - p) + 100(0.25) = **200p + 400**

EMV(d3) = 450p + 450(0.75 - p) + 400(0.25) = **437.5** (constant)

Check at p = 0.40: 425.0, 480.0, 437.5. Correct.

d3 is flat because the indoor arena pays the same 450 under both dry and mixed weather; shifting probability between two identical payoffs cannot change the expected value. Each slope equals that alternative's dry payoff minus its mixed payoff: 500, 200 and 0.

Comparison	Equation	Break-even p
d1 = d2	500p + 225 = 200p + 400 -> 300p = 175	0.5833
d1 = d3	500p + 225 = 437.5 -> 500p = 212.5	0.4250
d2 = d3	200p + 400 = 437.5 -> 200p = 37.5	0.1875

Strategy regions: 0 <= p < 0.1875 -> d3 (dry weather too unlikely to justify the open site) | 0.1875 < p < 0.5833 -> d2 (the compromise wins the middle range) | 0.5833 < p <= 0.75 -> d1 (dry weather likely enough to gamble on the 900).

The break-even p = 0.425 between d1 and d3 lies **inside** the region where d2 beats both, so it never appears as a boundary — exactly as the third indifference line behaves in the stock market example.

Robustness. The base case p = 0.40 sits 0.2125 above the lower boundary and 0.1833 below the upper one. The probability of a dry weekend would have to rise by more than 18 points or fall by more than 21 before the recommendation changed, so the choice of d2 is robust.

How much must d1's heavy-rain payoff improve? With x as that payoff, EMV(d1) = 0.40(900) + 0.35(400) + 0.25x = 500 + 0.25x. For d1 to overtake d2 we need 500 + 0.25x > 480, so **x > -80**: the worst case must be cut by 220.

Optional extension, EVPI. Best payoff per state is 900, 500, 400, so EVwPI = 0.40(900) + 0.35(500) + 0.25(400) = **635.0** and **EVPI = 635.0 - 480.0 = 155.0**. Perfect information changes the action in every state, which is why EVPI is large relative to the 42.5 gap between the best and second-best alternatives.

Part 5 — Worked Comprehensive Case 2: Factory Machine Investment

The contrasting case: here the probabilities are **act-dependent**.

Alternative	Upfront cost	Favourable	Prob.	Unfavourable	Prob.	EMV
d1 Buy new machines	200K	+300K	0.70	-100K	0.30	180K
d2 Keep old machines	0	+300K	0.60	+50K	0.40	200K (best)
d3 Lease refurbished	100K	+240K	0.55	+40K	0.45	150K

(a) Act-dependent probabilities. They differ across alternatives (0.70/0.30, 0.60/0.40, 0.55/0.45), so the distribution depends on the act chosen. Consequently there is no common set of states of nature: each alternative has its own chance node, and the influence diagram **requires an arc from the decision node to the chance node**. Cell-by-cell dominance testing across a single row of probabilities would not be legitimate here.

(c) Backward induction. EMV(d1) = 0.70(300) + 0.30(-100) = 210 - 30 = **180K** · EMV(d2) = 0.60(300) + 0.40(50) = 180 + 20 = **200K** · EMV(d3) = 0.55(240) + 0.45(40) = 132 + 18 = **150K**. Value of the decision node = **200K**; prune d1 and d3 with //.

(d) Cumulative risk profiles, F(x) = P(Payoff <= x)

x	F for d1	F for d2	F for d3
-100	0.30	0.00	0.00
40	0.30	0.00	0.45
50	0.30	0.40	0.45
240	0.30	0.40	1.00
300	1.00	1.00	1.00

P(Payoff <= 0): d1 = **0.30**, d2 = **0**, d3 = **0**. P(Payoff >= 300K): d1 = **0.70**, d2 = **0.60**, d3 = **0**.
Deterministic dominance: none — d2's worst (50) is below d3's best (240).
Stochastic dominance: d2 dominates d3, because F for d2 is at or below F for d3 at every x (0 <= 0.45, 0.40 <= 0.45, 0.40 <= 1.00). Eliminate d3.
d1 against d2 cannot be resolved: the curves cross — d2 is better in the lower tail, d1 in the upper.
Maximin gives d2 (50 against -100). **Maximax** ties at 300 and cannot discriminate.
Recommendation: risk neutral chooses d2 on EMV (200 against 180); risk averse also chooses d2, which removes the possibility of loss entirely. Both point to **keeping the old machines**.
(f) EVPI (the two uncertainties independent, d3 entering at its EMV of 150): EVwPI = 0.42(300) + 0.28(300) + 0.18(300) + 0.12(150) = **282K**, so **EVPI = 282 - 200 = 82K**. Excluding d3 gives EVwPI = 270 and EVPI = 70K — state whichever assumption you use.

Answer template for the comprehensive question — follow this order and label every part.
(a) Payoff matrix: alternatives by states, *net* payoffs, the probability on each cell. Add one sentence saying whether the probabilities are act-dependent or act-independent, and what follows from it.
(b) Influence diagram: list the nodes with their shapes, then list each arc and label it relevance or functional. State explicitly which arcs are *absent* and why. Note that there are no cycles and that the value node has no successors. Give the conditional probability table.
(c) Decision tree: write out every chance-node EMV in full, box the maximum at the decision node, mark losing branches with //, and finish with a sentence naming the optimal strategy and its EMV.
(d) Risk profiles, then the cumulative table F(x) = P(Payoff <= x), then three sentences in order: deterministic dominance, stochastic dominance and what is eliminated, and what a crossing means for what remains. Close with maximin, maximax and a recommendation for a risk-neutral and a risk-averse decision maker.
(e) Write each EMV as a linear function of p, set the pairs equal, solve for each break-even, state the strategy regions, and comment on whether the base case is robust.

Show every calculation. A correct number with no supporting work earns very little.